Course Introduction: AI and Applied Economics

Summer School in AI & Applied Economics

Elliott Ash

ETH Zürich

August 2026

Outline

Outline

Motivation: Why AI for Economists

Course Overview and Roadmap

A Brief Tour: From Text as Data to Automated Research

Welcome

- This course is about using **artificial intelligence** as a working tool in **applied economics and social science**.
- **Engineering goals:**
  - develop practical skills in machine learning, natural language processing, and agentic workflows;
  - apply them to real economic data — legal, political, media, and firm-level text and images.
- **Scientific goals:**
  - relate unstructured data to metadata in order to measure social and economic forces;
  - understand the decisions of judges, legislators, journalists, and firms through what they write and say.
- **Policy goals:**
  - ask what happens to labor markets, law, and science when machines can read and write.

Why now: the cost of reading collapsed

- Economists have always wanted to use documents as evidence. Until recently, reading them did not scale.
- Three things changed:
  1. **Supply of digitized data**: court opinions, legislative records, patents, filings, newspapers, parliamentary transcripts, satellite and street-level imagery.
  2. **Methods**: representation learning, transformers, instruction-tuned language models.
  3. **Cost**: annotating a million documents went from a research grant to an afternoon.

Three waves of text and AI in economics

1. **Dictionaries and bag-of-words** (roughly 2005–2015)
   - counts of chosen words; transparent, cheap, easily criticized;
   - e.g., Baker, Bloom, and Davis (QJE 2016) economic policy uncertainty.
2. **Learned representations** (roughly 2017–2022)
   - word2vec, GloVe, topic models, then contextual embeddings (BERT);
   - measurement of latent dimensions: ideology, slant, sentiment, culture.
3. **Large language models and agents** (2023–)
   - zero-shot annotation, extraction, summarization, and now code execution and multi-step research workflows.

- Each wave lowered the cost of *proposing* a measure. None removed the obligation to *validate* it.

Two distinct kinds of gains

- **Speed-ups**: things we already did, done faster and cheaper.
  - cleaning and linking messy administrative data;
  - coding a few thousand documents an RA would have coded;
  - writing and debugging estimation code;
  - drafting, translating, and literature search.
- **New methods**: research designs that were previously infeasible.
  - measuring dimensions with no closed-form dictionary (tone toward a group, legal ambiguity, argument structure);
  - corpus-scale designs: every judicial opinion, every parliamentary speech, every local newspaper;
  - automated hypothesis generation, simulation, and replication.

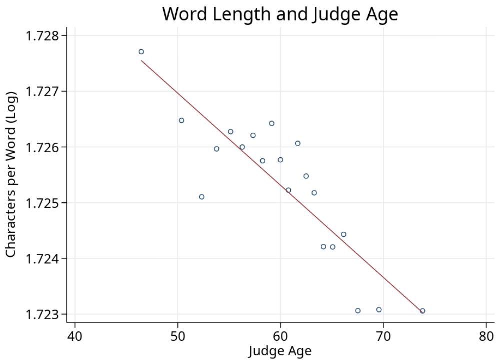
A useful discipline

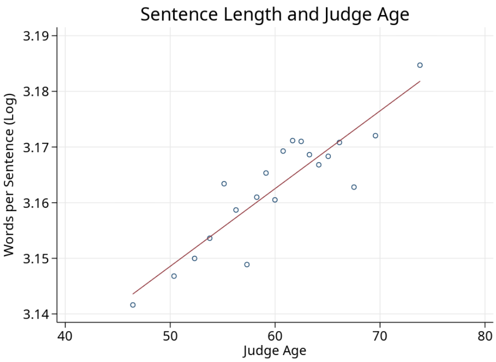
For every application: is this a speed-up, or a new method? The econometric issues are different.

The central caution: measurement error with structure

- An AI-generated variable $\hat{Y}$ is a *prediction* of a latent construct Y^*, not the construct itself.
- The error $\hat{Y} - Y^*$ is:
  - **not classical** — it correlates with the text, and hence with the treatment and the outcome;
  - **systematic across documents** — the same model makes the same mistake everywhere in your panel;
  - **invisible** — cheap to produce, expensive to audit.
- Consequences: attenuation, spurious significance, and confounding by writing style rather than substance.

Example: text reveals things metadata cannot

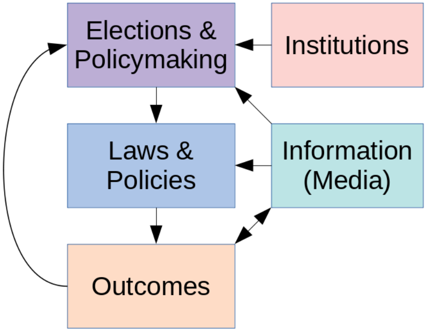




Documents sit inside a system, and the system is text

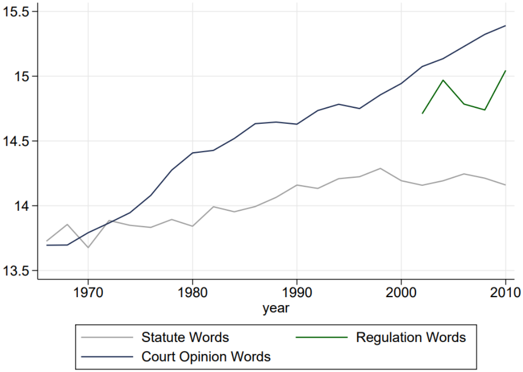
- **Institutions**: constitutions, charters, treaties
- **Elections and policymaking**: campaign ads, parliamentary debates, proposed bills
- **Media**: newspapers, TV transcripts, lobbying, academic research
- **Laws and policies**: legislation, regulation, judicial opinions
- **Outcomes**: contracts, culture, prices

Every node is observable as text. That is what makes the whole chain newly measurable.



Text corpora – have to think at a different scale

e.g., legal texts in U.S. states (Ash, Morelli, and Vannoni 2025)



Per year, U.S. states produce roughly:

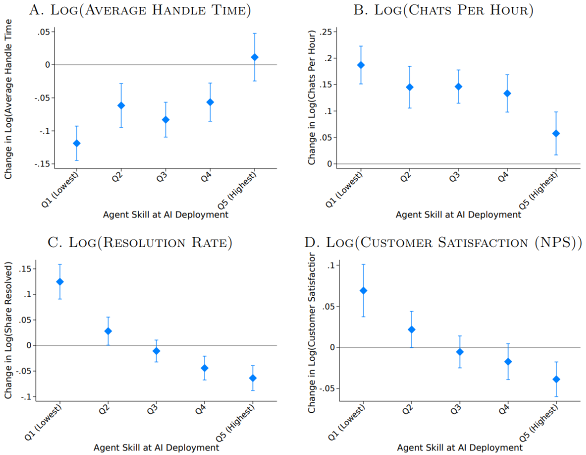
- ∼1.3M words of statutes
- ∼3.3M words of regulations
- ∼4.8M words of state court opinions

No research team reads this.
A model can.

AI is also an object of study

Brynjolfsson et al. (2025)

Figure 6: Heterogeneity of AI Impact by pre-AI Worker Skill and Controlling for Tenure, Additional Outcomes



Outline

Motivation: Why AI for Economists

Course Overview and Roadmap

A Brief Tour: From Text as Data to Automated Research

Structure of the Summer School

- **Week 1 — online (Zoom), 31 August – 4 September, afternoons CET**
  - eight lectures over four afternoons, plus a Friday recap-and-presentations session;
  - lectures pair concepts with hands-on notebooks.
- **Week 2 — in person at ETH Zürich (RZ building, room F21), 7–10 September**
  - faculty research sessions each morning;
  - project studio and advisor meetings each afternoon.

Week 1 roadmap (1/2)

Monday

1. **Course introduction** — AI and applied economics *(today, Ash)*
2. **ML basics, embeddings, and clustering** — prediction, regularization, validation; representations and unsupervised structure *(Ciccarone)*
3. **Text and image data** — text and images as economic measurement *(Ciccarone)*

Tuesday

4. **ML and econometrics** — inference with machine-generated variables *(Gauthier)*
5. **LLMs and alignment** — how a language model works (tokens, attention, transformers), then instruction tuning, RLHF, reasoning, RAG, and agents *(Ash)*

Week 1 roadmap (2/2)

Wednesday

6. **Working with AI agents** — orchestration, tooling, evaluation *(Roeben)*
7. **Software and application development with AI** — building research infrastructure *(Roeben)*

Thursday

8. **Training LLMs** — running and adapting models in practice: APIs, open weights, fine-tuning, LoRA *(Drinkall)*

Friday: Week 1 recap, conclusion *(Ash)*

- The arc: *measure* → *model* → *infer* → *automate*.

Week 2: research sessions in Zürich

We will see:

In applied AI/econ papers, the model choice is usually not that interesting. Measurement validation, research design, and scientific contribution are paramount.

Assignments

- **Week 1: problem set and web app.**
  - analyze text and images from the supplied Fox/MSNBC YouTube dataset or a dataset of your choice;
  - on Day 3, build a web app to explore or visualize the analysis;
  - present the web app during Friday's recap session.
- **Week 2: project development.**
  - work on your own project or join a group project;
  - use AI as a research method, study AI as an object, or use AI to build research infrastructure;
  - make concrete progress in project-studio and advisor sessions.

Preparation and tools

- Bring a laptop:
  1. access to a capable **AI service** — ChatGPT Plus or Claude Pro (or higher); contact the organizers if this is not feasible;
  2. at least one **coding agent** installed: Codex and/or Claude Code;
  3. **Git** and a working **Python/Jupyter** environment.

- Notebooks and data links are distributed in the course repository.
- Corpora: get familiar with Hugging Face Datasets (`https://huggingface.co/docs/datasets/`), REST APIs, and basic scraping.

How to get the most out of this course

- **Bring a question, not a method.** The best projects here start from an economic question that a corpus can answer.
- **Run and refine the notebooks, don't just read them.** The friction is the lesson.
- **Break the models on purpose.** Feed them adversarial documents and see where the measure fails.
- **Validate everything you measure** against human labels, even a hand-coded sample of 100 documents.
- **Use the agents openly.** Tell us when they go wrong.

Outline

Motivation: Why AI for Economists

Course Overview and Roadmap

A Brief Tour: From Text as Data to Automated Research

What is the endpoint of Natural Language Processing (NLP)?

Machine understanding of **discourse** across long documents and corpora.

- good summaries of long texts: extract the relevant, discard the irrelevant;
- question answering: retrieve evidence and answers from large corpora;
- reasoning over a body of documents that no human could read.

- Are we there now?
- If not, what is missing — capability, evaluation, or data?

Four modes for NLP

- **Local**: linguistic information from local context (sentences, paragraphs)
  - local sentiment, textual entailment, co-reference resolution, closed question answering.
- **Long document** (covering multiple topics)
  - TF-IDF and bag-of-words representations $\rightarrow$ supervised learning; cosine distance between document vectors.
- **Global / knowledge base**: corpus-level tasks
  - information retrieval and search; open question answering; claim checking.
- **Generative / creative**: produce text for a purpose
  - draft a summary, a legal brief, a counterargument, or a simulated survey response.

Objectives: social-science research using text data

1. **What is the research question?**
2. **Corpus and data**
   - obtain, clean, preprocess, and link to metadata;
   - produce descriptive statistics and visuals on text and metadata.
3. **Language modeling**
   - **what exactly are we trying to measure?**
   - select a model and train (or prompt) it;
   - probe sensitivity to hyperparameters and prompts;
   - **validate** that it measures what you claim.
4. **Empirical analysis**
   - produce statistics or predictions with the trained model;
   - **answer the research question.**

Text is high-dimensional

- Text data is a sequence of characters called **documents**; the set of documents is the **corpus** D.
- For documents of fixed length n_L drawn from a vocabulary of size n_V, the number of distinct possible documents is $n_V^{n_L}$.
  - 30-word messages using only the 1,000 most common English words: $(10^3)^{30} = 10^{90}$ distinct documents.
- Represent a document as a matrix of n_V-dimensional categorical variables:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_t & \cdots & \mathbf{x}_{n_L} \end{bmatrix}$$

  - dimensionality of one document $= n_L n_V$, sparse with n_L non-zero entries;
  - for $n_L = 30$, $n_V = 1{,}000$: dimensionality $= 30{,}000$.

Text data is unstructured

- The information we want is mixed together with (a lot of) information we do not want.
- **Every** text-as-data approach throws away information. The trick is choosing what to keep.
- Tokenization and dimension reduction are exactly this step: transforming an unstructured corpus D into a usable matrix X.

The same logic applies to LLMs

Prompting a model to output a number is also a dimension-reduction choice — it is just made implicitly, by the model, rather than explicitly, by you.

Addressing high dimensionality in text

Ash & Hansen (2023)

- **Dictionaries**: count the terms you care about (Baker et al. 2016; Hassan et al. 2019; Enke 2020)
  $\rightarrow$ dimensionality is the number of tagged categories.
- **Bag-of-words models**: count each word/phrase in a vocabulary (Hoberg & Phillips 2016; Cagé et al. 2020; Kelly et al. 2021)
  $\rightarrow$ dimensionality is the vocabulary size.
- **Factor models**: apply PCA or LDA (topic models) to the bag-of-words representation (Hansen et al. 2018; Bertrand et al. 2021)
  $\rightarrow$ dimensionality is the number of low-dimensional factors.
- **Static word embeddings**: vectorize documents as averaged word embeddings (Gennaro & Ash 2022; Ash et al. 2024)
  $\rightarrow$ dimensionality is determined by the embeddings.
- **Sequence models (LLMs)**: process the token sequence itself
  $\rightarrow$ Tuesday's lecture.

Why context matters: what counts cannot see

- **Synonymy** — different words, same meaning:
  - the legal ruling was *rejected*
  - the legal ruling was *invalidated*
- **Polysemy** — same word, different meanings:
  - The court *stayed* the proceedings.
  - The defendant *stayed* in the jurisdiction.
- **Sequence** — same words, different meaning:
  - Congress passed *limits* on the budget and *expansions* on taxes.
  - Congress passed *expansions* on the budget and *limits* on taxes.

- Counts miss all three. Models that read words *in context* — the sequence models above — can address these limitations.

This course is about relating documents to metadata

- Applied NLP: the documents are rarely interesting on their own.
- We want to relate **text** to **metadata**.
- e.g., positive–negative sentiment Y in judicial opinions is not meaningful by itself.
- But sentiment Y_{ijt} in opinion i, by judge j, at time t:
  - how does it vary over time t?
  - does a judge from party p_j express more negative sentiment toward defendants from group g_i?

Rule of thumb

A text measure becomes economics only when it is indexed by a unit, a time, and a source of variation.

Dictionary methods: still a workhorse

- A pre-selected list of words or phrases, counted across documents.
  - e.g., how often a judge says “justice” versus “efficiency”.
- Why economists keep using them despite obvious limitations:
  - transparent and reproducible;
  - the researcher explicitly “regularizes out” confounding language dimensions rather than letting a model absorb them.
- Limitation: brittle to negation, domain shift, and sarcasm — and the word list is itself a researcher degree of freedom.

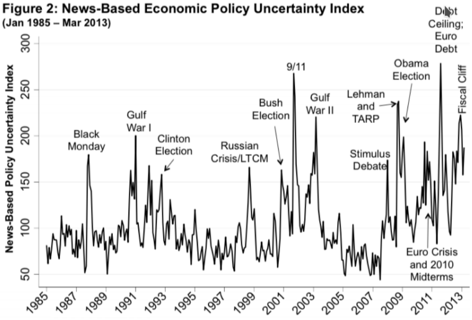
Dictionary methods in practice: policy uncertainty

Baker, Bloom, and Davis (QJE 2016)

For each newspaper on each day since 1985, count articles that contain:

1. "uncertain" OR "uncertainty"; **and**
2. "economic" OR "economy"; **and**
3. "congress" OR "deficit" OR "federal reserve" OR "legislation" OR "regulation" OR "white house".

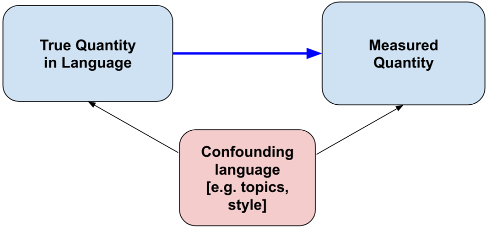
Normalize by total articles that month.



How would you do this with an LLM?

Supervised measures and their confounders

- Off-the-shelf sentiment models fail on domain text: in a court opinion, "murder" scores negative regardless of the judge's tone.
- More generally:
  - **supervised** models learn features correlated with the annotated label, not only its true cause;
  - **unsupervised** models learn correlations between topics and contexts;
  - **Contextual language models**, including BERT-type encoders, inherit both problems and can absorb additional language confounders.



LLMs: From measurement tool to research collaborator

- First-generation use: LLM as an **annotator** — classify, extract, summarize.
- Second-generation use: LLM as an **agent** — read a paper, write code, run the analysis, inspect the output, revise.
- This changes which parts of the research process are scarce.
  - Cheap: implementation, literature search, robustness checks.
  - Still scarce: question selection, identification, validation, and judgment about what is interesting.

“Prompt engineering” for research

Prompting Science Report 1: Prompt Engineering is Complicated and Contingent

10 Pages
Posted: 5 May 2025

Lennart Meincke
University of Pennsylvania; The Wharton School; WHU - Otto Beisheim School of Management

Ethan R. Mollick
University of Pennsylvania - Wharton School

Lilach Mollick
University of Pennsylvania - Wharton School

Dan Shapiro
Glowforge, Inc; University of Pennsylvania - The Wharton School

Date Written: March 04, 2025

Abstract

This is the first of a series of short reports that seek to help business, education, and policy leaders understand the technical details of working with AI through rigorous testing. In this report, we demonstrate two things:

- There is no single standard for measuring whether a Large Language Model (LLM) passes a benchmark, and that choosing a standard has a big impact on how well the LLM does on that benchmark. The standard you choose will depend on your goals for using an LLM in a particular case.

- It is hard to know in advance whether a particular prompting approach will help or harm the LLM's ability to answer any particular question. Specifically, we find that sometimes being polite to the LLM helps performance, and sometimes it lowers performance. We also find that constraining the AI’s answers helps performance in some cases, though it may lower performance in other cases.

- Treat the prompt as a **codebook**: written so that a human coder and the model would produce the same label.

Using LLM outputs in regressions

Using Large Language Model Annotations for the Social Sciences:
A General Framework of Using Predicted Variables
in Downstream Analyses*

Naoki Egami[†] Musashi Hinck[‡] Brandon M. Stewart[§] Hanying Wei[¶]

First Version: May 13, 2024
This Version: November 17, 2024

Abstract

Social scientists use automated annotation methods, such as supervised machine learning and, more recently, large language models (LLMs), that can predict labels and generate text-based variables. While such predicted text-based variables are often analyzed as if they were observed without errors, we show that ignoring prediction errors in the automated annotation step leads to substantial bias and invalid confidence intervals in downstream analyses, even if the accuracy of the automated annotations is high, e.g., above 90%. We propose a framework of *design-based supervised learning* (DSL) that can provide valid statistical estimates, even when predicted variables contain non-random prediction errors. DSL employs a doubly robust procedure to combine predicted labels and a smaller number of expert annotations. DSL allows scholars to apply advances in LLMs to social science research while maintaining statistical validity. We illustrate its general applicability using two applications where the outcome and independent variables are text-based.

Inference for Regression with Variables Generated by AI or Machine Learning*

Laura Battaglia[†] Timothy Christensen[‡]
Stephen Hansen[§] Szymon Sacher[¶]

April 30, 2025

Abstract

Researchers now routinely use AI or other machine learning methods to estimate latent variables of economic interest, then plug-in the estimates as covariates in a regression. We show both theoretically and empirically that naively treating AI/ML-generated variables as "data" leads to biased estimates and invalid inference. To restore valid inference, we propose two methods: (1) an explicit bias correction with bias-corrected confidence intervals, and (2) joint estimation of the regression parameters and latent variables. We illustrate these ideas through applications involving label imputation, dimensionality reduction, and index construction via classification and aggregation.

- Practical guidance: hand-label a validation sample; report error rates by treatment group; show how estimates move under measurement-error corrections.

Case study: can agents replicate published research?

Kohler et al., “Read the Paper, Write the Code” (2026)

- AI models can now write and execute code autonomously.
- Social science journals increasingly mandate replication packages (data + code) — yet these packages are rarely re-run, and the reproducibility crisis persists.

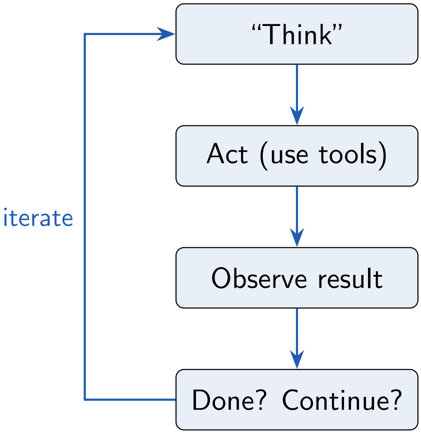
This paper asks:

Can AI agents automatically replicate empirical social science research, given only the paper and the raw data?

What Is an AI Agent?

A simple definition

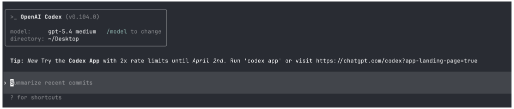
- Reasoning LLM +
- Chained Requests +
- Tools
  - Open files and read data
  - Write and run code
  - See errors, debug, and retry
  - Work autonomously for a long time



"Coding" Agents in Practice

Claude Code (Anthropic)

Codex (OpenAI)



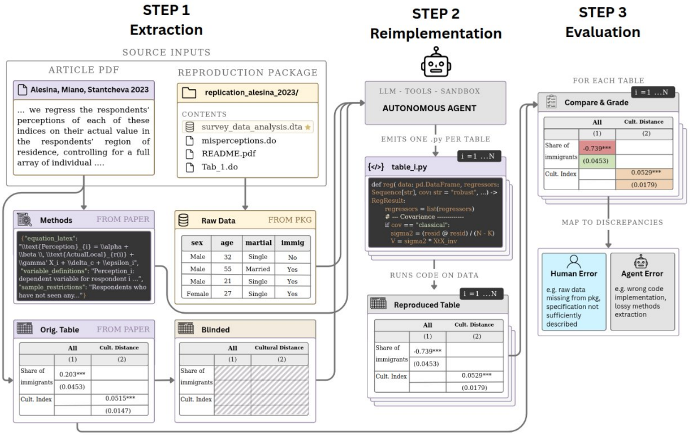


Figure 1. Overview of the pipeline for replicating empirical results. *Step 1 (Extraction): Paper and replication package are parsed to extract the methods, original data, and a blinded version of the original results table. Step 2 (Reimplementation): An autonomous LLM agent, equipped with tools and a sandboxed execution environment, emits one Python script per table and runs it on the original data to produce a reproduced table. Step 3 (Evaluation): Each reproduced table is compared cell-by-cell against the original, and divergences are traced back to human errors (e.g., missing data, underspecified methods in the original package) and agent errors (e.g., incorrect code, lossy extraction of the methods).*

The testbed: 48 papers, 14,214 blank cells to fill

- 48 papers, top econ & poli-sci journals (AJPS, EJ, AER, JOP, . . .), all human-verified reproducible by **I4Replication**
- Extracted **222 tables** / **14,214 cells** (5,149 coefficients, 4,253 standard errors)
- Task: fill the **redacted** cells from methods + data alone

Panel A: Papers by journal and discipline (final sample, $N = 48$)

Journal	Discipline	N papers
American Journal of Political Science (AJPS)	Political Science	11
Economic Journal (EJ)	Economics	9
American Economic Review (AER)	Economics	8
American Economic Journal: Economic Policy (AEJ:Pol)	Economics	5
Journal of Politics (JOP)	Political Science	5
American Economic Journal: Applied Economics (AEJ:AE)	Economics	3
American Political Science Review (APSR)	Political Science	3
Review of Economic Studies (REStud)	Economics	2
American Economic Journal: Macroeconomics (AEJ:Macro)	Economics	1
Quarterly Journal of Economics (QJE)	Economics	1
Total		48

Panel B: Extracted Elements in Original Papers (per paper, $N = 48$)

	Total	Mean	SD	Min	Max
Tables	**222**	4.6	2.8	1	11
Cells (present)	**14,214**	296.1	261.4	20	1433
Coefficients	**5,149**	107.3	93.5	8	384
Standard errors	**4,253**	88.6	85.3	0	319
p-values	**779**	16.2	32.5	0	127
t-statistics	**10**	0.2	1.4	0	10
Confidence intervals	**112**	2.3	16.2	0	112
R-squared	**590**	12.3	17.9	0	58
N observations	**1,701**	35.4	86.2	0	602
F-statistics	**121**	2.5	6.2	0	30
Other numeric	**1,607**	33.5	53.6	0	253

Originals are Stata/R — agents rewrite in Python

Original replication packages:

- **54%** Stata, **27%** R, **2%** MATLAB, **0%** Python
- avg **5,324** lines of code

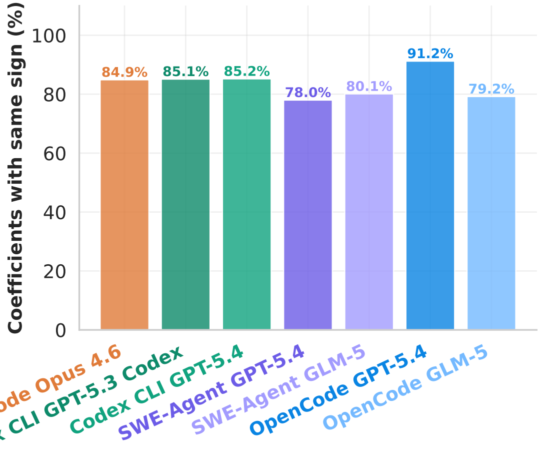
Language	N papers	% papers	Total LOC	Mean LOC
Stata	26	54.2%	203,953	7,844
R	13	27.1%	24,462	1,881
MATLAB	1	2.1%	5,520	5,520
mixed	7	14.6%	21,637	3,091
All	48	100.0%	255,572	5,324

Table A2. Primary language of replication packages in the final sample. LOC is a line-count across all code files.

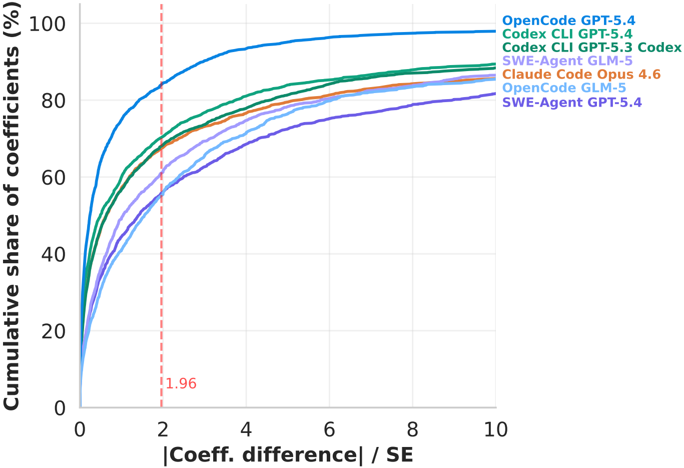
Agents rewrote everything in Python — this is reimplementation, not re-running code.

Also weakens any leakage worry: memorized results would have to be translated *across languages*.

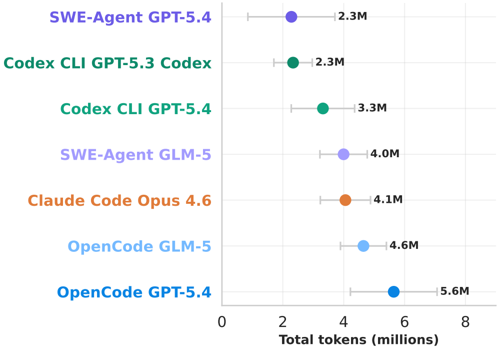
Cell-level results: correct coefficient sign

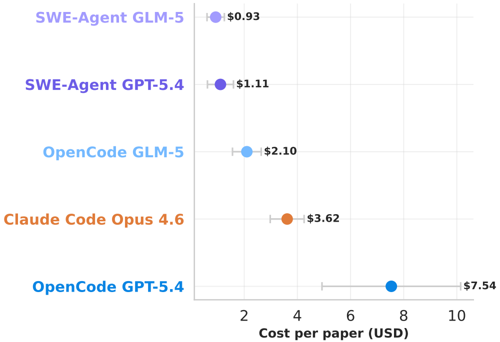


Coeff. Match: CDF of Statistical Difference (β/SE) from Original

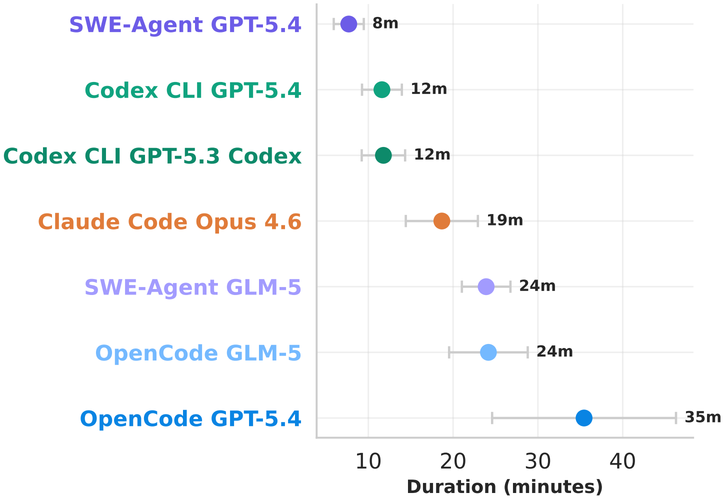


Token Usage and Costs





Processing time



Bonus: Replicating Plots

Volume 84 Number 3 July 2022 / 1831

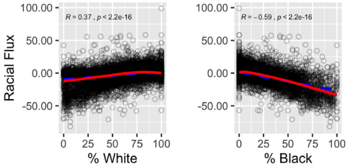
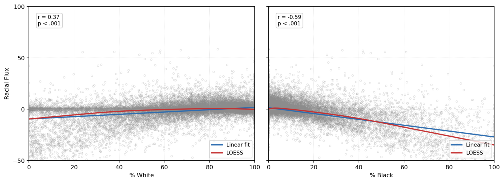


Figure 1. Racial flux and residential racial composition

Original

Figure 1. Racial flux and residential racial composition



Reproduced

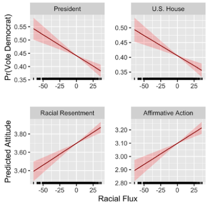
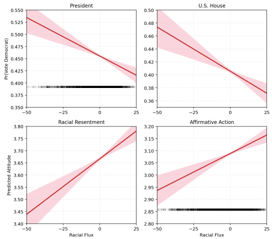


Figure 2. Racial flux, voting behavior, and racial attitudes (Whites)—predicted probabilities.

Figure 2. Racial flux, voting behavior, and racial attitudes (Whites) - predicted probabilities



The surprise: failures mostly aren't the AI's fault

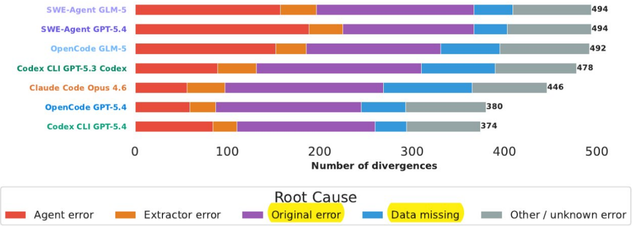


Figure 7. Error analysis pipeline: Number of divergences by error source. *The error pipeline identifies divergences between the original and reproduced code for tables that score worse than overall grade B. These divergences are described and classified into categories. The agent then traces the root cause of the divergence.* ***Agent error:*** *the agent makes an error in mapping the methods to reproduced code.* ***Extractor error:*** *the initial extraction from paper to methods description is faulty.* ***Original error****: the paper under-specifies or mis-specified the original code.* ***Data missing****: data is missing in the reproduction package. Appendix A.7 has more detail.*

Over **3/4** of divergences trace to a specific source:

$\sim$**50%** paper underspecification

$\sim$**16%** missing data

$\approx$**66%** human-attributable

For the best agent, agent mistakes are just $\sim$**8%**.

Underspecification, concretely

Paper says *"controls included"*; the code actually uses specific variables, filters, transformations. The agent must **guess** — and different agents guess differently.

	10.1093/ej/ueab069 — Table 3a Behavior	Gr.	10.1257/pol.20200559 — Table 2 Behavior	Gr.	10.1093/ej/ueab102 — Table 3 Behavior	Gr.
Claude code behavior	Agent assumes id for democrats is party=100	C	Agent assumes that codes A1 and A2 indicate democrat support	D	Uses `linearmodels'` `diagnostics.f.stat`	D
OpenCode 5.4 behavior	Agent assumes id for democrats is party=200	B	Agent assumes that codes A4 and A5 indicate democrat support	B	Manually reconstructs the (Kleibergen-Paap) Wald F-statistic	B
Correct behavior	Democrats id is party=200		A4 and A5 indicates democrat support		Stata uses `ivreg2` (Kleibergen-Paap) `widstat` F-statistic	
Error Attribution	Paper underspecifies code by not disclosing how party is coded in data		Paper underspecifies code by not disclosing how party is coded in data		Paper underspecifies F-statistic type	

Table 3. Examples of discrepancies of paper underspecification. *The examples illustrate how agent behavior can vary with an underspecified task.*

The bottleneck is shifting

6 Conclusion

This paper establishes that AI agents can, in many cases, recover social science results from a paper's text and data alone. Across a diverse set of empirical papers, agents are able to reconstruct a large share of published results without access to the original code, demonstrating substantial progress toward automated reproducibility. While this first generation of agents delivers imperfect results, the overall failure modes reveal a clear pattern. Going forward, the primary bottleneck may not be model capability, but the way social science methods are documented. In many cases, discrepancies arise because key implementation details are underspecified or omitted entirely, forcing agents to infer choices (just as a human researcher would under the same constraints). As a result, the ability to reproduce results reflects a combination of both the agent's capabilities and the paper's clarity and completeness.

Not model capability → **documentation quality**.

If papers aren't the source of truth, what are they for?

We began this paper by articulating the common view that the scientific paper is the source of truth for scientific claims. However, many of the discrepancies we observe arise from missing or ambiguous specifications, such as variable coding choices, data filters, or estimation procedures. In this light, it is worth reconsidering what we mean by the "source of truth." For the purpose of reproducing results, that role is already fulfilled by the code, which provides a complete, machine-readable specification of the analysis pipeline and resolves ambiguities that the paper often leaves implicit. This suggests a clearer division of roles: code serves as the authoritative representation of *what was done*, while the paper explains *why* those choices were made, defining variables, motivating identification strategies, and interpreting results in context.

Code defines *what was done*. **Paper** explains *why*.

Reproduction agents as a diagnostic

This distinction has practical implications for how reproducibility is evaluated. Rather than expecting papers to fully encode implementation details, a more robust approach is to require explicit alignment between the narrative and the underlying code. Agentic systems provide a natural mechanism for this: they can translate papers into executable code, compare outputs, and surface inconsistencies between the two representations. Thus, automated reproduction can serve as both a tool for verification and a diagnostic, identifying where methods are underspecified, where implementations diverge, and where assumptions must be inferred. Aligning research practices with the requirements of both human and automated interpretation—through more explicit, structured, and complete method descriptions—may therefore play an important role in improving reproducibility at scale.

Automated reproduction can **verify paper/code alignment** — surfacing where methods are underspecified, implementations diverge, or assumptions must be inferred.

Implications for how we do research

For social science practice:

- methods sections alone are frequently *insufficient* to reproduce an analysis;
- structured, machine-readable methods summaries may become part of submission;
- automated reproducibility checks are now technically feasible.

For AI agents:

- scaffold–model alignment matters as much as raw model quality;
- the frontier moves fast enough that results date within months;
- error rates would fall substantially with better *paper* practices.

Direction of travel

A first step toward autonomous *conceptual* reproduction — from data collection through method design.

AI and (researcher) human capital

- AI can *reduce the cost of learning*: an always-available tutor that explains concepts on demand.
- AI can also *substitute for the learning process*: the student completes the task without acquiring the skill.

Relevance to this course

We assume you use AI throughout. The skill we care about is not “can you write the code” but “can you tell when the output is wrong”.

Terence Tao: *"My suggested rule of thumb: if the authors cannot convincingly demonstrate that they can give a clear, expert-level talk on their results, that is correct and properly attributed, then the result should not be published."*

Caleb Watney @calebwatney

This is the best paper written so far about the impact of AI on scientific discovery

Artificial Intelligence, Scientific Discovery, and Product Innovation*

Aidan Toner-Rodgers†
MIT

November 6, 2024

This paper studies the impact of artificial intelligence on innovation, exploiting the randomized introduction of a new materials discovery technology to 1,018 scientists in the R&D lab of a large U.S. firm. AI-assisted researchers discover 44% more materials, resulting in a 39% increase in patent filings and a 17% rise in downstream product innovation. These compounds possess more novel chemical structures and lead to more radical inventions. However, the technology has strikingly disparate effects across the productivity distribution: while the bottom third of scientists see little benefit, the output of top researchers nearly doubles. Investigating the mechanisms behind these results, I show that AI automates 57% of "idea-generation" tasks, reallocating researchers to the new task of evaluating model-produced candidate materials. Top scientists leverage their domain knowledge to prioritize promising AI suggestions, while others waste significant resources testing false positives. Together, these findings demonstrate the potential of AI-augmented research and highlight the complementarity between algorithms and expertise in the innovative process. Survey evidence reveals that these gains come at a cost, however, as 82% of scientists report reduced satisfaction with their work due to decreased creativity and skill underutilization.

Caleb Watney @calebwatney

Woof. Very sad to see. The MIT econ department has recommended that this paper be withdrawn...

MIT Economics

Assuring an accurate research record

May 16th, 2025

Following the posting of the preprint paper "Artificial Intelligence, Scientific Discovery, and Product Innovation" on arXiv in November 2024, concerns were raised about the integrity of the research. MIT conducted an internal, confidential review and concluded that the paper should be withdrawn from public discourse.

What do human researchers still do?

- **Choose the question.** Models optimize objectives; they do not decide what is worth knowing.
- **Own the identifying assumption.** No amount of generated text makes an exclusion restriction credible.
- **Validate the measure** against ground truth the model has never seen.
- **Take responsibility.** Authorship means accountability for every number in the paper.

The bet behind this Summer School

The returns to economic judgment go *up*, not down, as implementation becomes free — provided you understand the tools well enough to supervise them.

Where we go from here

- Next, today: machine learning foundations, embeddings and clustering, then text and image data.
- Tuesday: econometrics with machine-generated variables; then how language models work and what alignment does to them.
- Wednesday: working with agents; building research software with AI.
- Thursday: training and adapting LLMs in practice.
- Week 2: faculty research sessions and your own project.

Questions and comments?